

Homogeneity

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In First-Order ODEs (Function Scaling)

A first-order ODE in the form: $\frac{dy}{dx} = f(x, y)$ is considered homogeneous if the function $f(x, y)$ depends on the ratio $\frac{y}{x}$.

To test for homogeneity:

$$f(tx, ty) = t^n f(x, y)$$

A function is homogeneous of degree n if both variables are able to be scaled by constant t and yields the original function multiplied by t^n .

If the function is rewritten in:

$$M(x, y)dx + N(x, y)dy = 0$$

The function is homogeneous if the degree of $M(x, y)$ and $N(x, y)$ are equal.

For example:

$$(x^2 + y^2)dx + xydy = 0$$

This function is homogeneous.

Individually, x^2 and y^2 both have degree of 2.

xy also has degree of 2. As $x^1 \times y^1 = xy$.

EVERY TERM IS DEGREE 2.

- If the degrees are mixed, that is not homogeneous.
- You must also inspect the variables.
- If the variables x and y always appear to be grouped together as a fraction, the function is homogeneous.

Examples:

- $\frac{x}{y}$
- $\frac{y}{x}$
- $\sin\left(\frac{x}{y}\right)$

Homogeneous first-order equations cannot have standalone numbers added or subtracted. This is because constants break the scaling property.

$\frac{dy}{dx} = \frac{x+y}{x}$ is homogeneous.

$\frac{dy}{dx} = \frac{x+y+1}{x}$ is not homogeneous.

Solving First-Order ODEs

The idea to solve any first-order ODE is to simplify it by substituting the ratio of x and y .

$$v = \frac{y}{x} \rightarrow y = vx$$

Remember that:

$$v = \frac{y}{x} \tag{1}$$

$$y = vx \tag{2}$$

$$\frac{dy}{dx} = v + x \frac{dv}{dx} \tag{3}$$

Substitute these values into the original equation.

Then to apply separation of variables. Before converting v back to the ratio.

Homogeneity in Linear ODEs (Structure and Form)

A linear ODE is homogeneous if every term contains the dependent variable or one of its derivatives AND if the equation is set to 0.

It can generally be written as:

$$a_n(x)y^n + a_{n-1}(x)y^{n-1} + \dots + a_1(x)y' + a_0(x)y = 0$$

There can be no forcing term.

If $y = 0$ is a valid solution, it is homogeneous.

If y_1 and y_2 are solutions to a linear homogeneous ODE, then any linear combination of $C_1y_1 + C_2y_2$ is also a solution.

To solve

If the coefficients are constant, use a characteristic polynomial.